BLOCKCHAIN ENGINEERING

ASSIGNMENT 12

Problem Statement

Implement a basic blockchain-based portfolio system where a user can:

• Create assets (e.g., certificates, projects, or tokens)

• View all assets stored on the blockchain

• Transfer asset ownership between users

This practical demonstrates how Hyperledger Fabric solves real- world problems like data immutability, transparency, and secure ownership transfer.

Requirements

Software Requirements

• Operating System: Windows / Linux / macOS

• Node.js (v16 or later)

• Docker & Docker Compose

• Git

• VS Code (with Docker and JavaScript extensions)

STEP 1: create folder

```
PS C:\Users\ADMIN> cd desktop
PS C:\Users\ADMIN\desktop> mkdir blockchain-project
>> cd blockchain-project


    Directory: C:\Users\ADMIN\desktop


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----        16-04-2026     10:04                blockchain-project
```

```
PS C:\Users\ADMIN\desktop\blockchain-project> New-Item index.html
>> code index.html


    Directory: C:\Users\ADMIN\desktop\blockchain-project


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        16-04-2026     10:23              0 index.html


PS C:\Users\ADMIN\desktop\blockchain-project>
                                                 start index.html
PS C:\Users\ADMIN\desktop\blockchain-project>
```

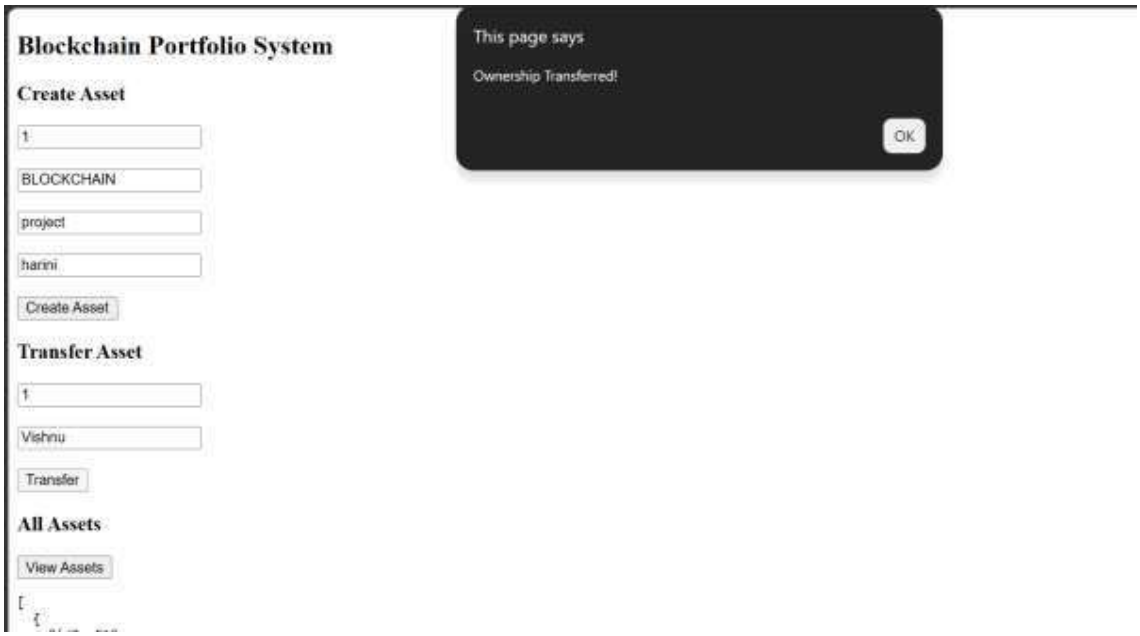
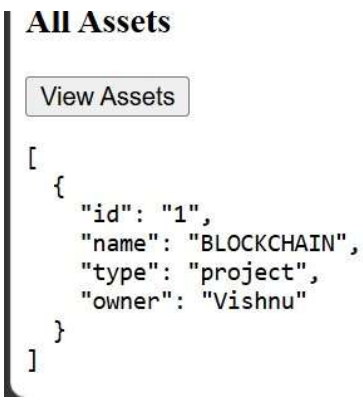
CODE:

```
C: > Users > ADMIN > desktop > blockchain-project > index.html > html
<!DOCTYPE html>
<html>
<head>
    <title>Blockchain Portfolio</title>
</head>
<body>

<h2>Blockchain Portfolio System</h2>

<h3>Create Asset</h3>
<input id="id" placeholder="Asset ID"><br><br>
<input id="name" placeholder="Name"><br><br>
<input id="type" placeholder="Type"><br><br>
<input id="owner" placeholder="Owner"><br><br>
<button onclick="createAsset()">Create Asset</button>

<h3>Transfer Asset</h3>
<input id="tid" placeholder="Asset ID"><br><br>
<input id="newOwner" placeholder="New Owner"><br><br>
<button onclick="transferAsset()">Transfer</button>

<h3>All Assets</h3>
<button onclick="viewAssets()">View Assets</button>
```

```
C: > Users > ADMIN > desktop > blockchain-project > index.html > html > body
<html>
<body>
<pre id="output"></pre>

<script>
let assets = [];

function createAsset() {
    let asset = {
        id: document.getElementById("id").value,
        name: document.getElementById("name").value,
        type: document.getElementById("type").value,
        owner: document.getElementById("owner").value
    };
    assets.push(asset);
    alert("Asset Created!");
}

function viewAssets() {
    document.getElementById("output").textContent = JSON.stringify(assets, null, 2);
}

function transferAsset() {
    let id = document.getElementById("tid").value;
```

```
C: > Users > ADMIN > desktop > blockchain-project > index.html > html > body
<html>
<body>
<script>
function transferAsset() {
    let newOwner = document.getElementById("newOwner").value;

    for (let i = 0; i < assets.length; i++) {
        if (assets[i].id === id) {
            assets[i].owner = newOwner;
            alert("Ownership Transferred!");
            return;
        }
    }
    alert("Asset Not Found");
}
</script>

</body>
</html>
```

OUTPUT :

STEP1 : CREATE ASSRT

STEP2: TRANSFER ASSET